

BEFORE	AFTER	file
59 int i, j, k;	59 int i, j, k;	
60	60	
61 #pragma scop	61 #pragma scop	
62 for (k = 0; k < _PB_N; k++)	62 for (k = 0; k < _PB_N; k++)	
63 {	63 {	
	64 /* Normalize row k: each j normalizes A[k][j] independently */	
	65 #pragma omp parallel for	
64 for (j = k + 1; j < _PB_N; j++)	66 for (j = k + 1; j < _PB_N; j++)	
65 A[k][j] = A[k][j] / A[k][k];	67 A[k][j] = A[k][j] / A[k][k];	
	68	
	69 /* Update submatrix: each (i,j) pair updates A[i][j] independently.	
	70 k is loop-invariant, reads from row k and column k (already computed)	
	71 Using collapse(2) to parallelize the full 2D region of independent up	
	72 #pragma omp parallel for collapse(2)	
66 for(i = k + 1; i < _PB_N; i++)	73 for(i = k + 1; i < _PB_N; i++)	
67 for (j = k + 1; j < _PB_N; j++)	74 for (j = k + 1; j < _PB_N; j++)	
68 A[i][j] = A[i][j] - A[i][k] * A[k][j];	75 A[i][j] = A[i][j] - A[i][k] * A[k][j];	
69 }	76 }	
70 #pragma endscop	77 #pragma endscop	